ACMFNet: Attention-Based Cross-Modal Fusion Network for Building Extraction of Remote Sensing Images

Baiyu Chen, Zongxu Pan, *Senior Member, IEEE*, Jianwei Yang, and Hui Long

***Abstract*— In recent years, significant progress has been made in extracting buildings from high spatial resolution (HSR) remote sensing images due to the rapid development of deep learning (DL). However, the existing methods still have some limitations in maintaining the detail integrity of building footprint. First, skip connections typically involve the direct concatenation of feature maps from adjacent levels, which inevitably leads to misalignment due to semantic differences. Second, the integration of building-related details remains a challenging task in the context of cross-modal remote sensing image. Third, the over-simplified upsampling structure used in previous methods may lead to loss of spatial details. In this article, we propose a novel building extraction method attention-based cross-modal fusion network (ACMFNet) based on cross-modal HSR remote sensing images using an encoder–decoder structure. First, we propose a global and local feature refinement module (GL-FRM) to refine features and establish contextual dependencies at multiple scales and levels, mitigating the spatial discrepancy among multilevel features. Meanwhile, a cross-modal fusion module is utilized to integrate complementary features extracted from multispectral (MS) data and normalized digital surface model (nDSM) data. In addition, we employed a lightweight residual upsampling module (RUM) for feature resolution recovery. We conducted complete experiments on two benchmark datasets, and the results indicate that our proposed ACMFNet achieves state-of-the-art (SOTA) performance without bells and whistles.**

***Index Terms*— Building extraction, cross-modal fusion, digital surface model, high spatial resolution (HSR), remote sensing image.**

I. Introduction

With the rapid advancement of aerospace and sensor technology, researchers can easily acquire a substantial amount of high spatial resolution (HSR) remote sensing images that depict the ecological environment and human activities [1]. Currently, the semantic segmentation task developed in the computer vision community has gained significant attention in the field of remote sensing [2], [3], [4]. Building extraction from HSR remote sensing images could be classified under this task, aiming to label all pixels in an image as either building or nonbuilding classes [5], which plays a pivotal role in applications, such as urban planning [6], [7], [8], population statistics [9], economic assessment [10], and disaster management [11], [12].

Since entering the deep learning (DL) era, as a mainstream framework for automatic building extraction, full convolutional network (FCN) [13] transforms the general image segmentation task into a pixel-level classification task. Based on the framework of FCN, a series of methods for automatic building extraction have been proposed [14], [15], [16]. However, despite the excellent progress of DL-based building extraction methods, there are still some persistent problems in the current remote sensing building extraction field.

A. Invalid Propagation of Semantics and Inefficient Self-Attention (SA) Computation Strategy

The majority of DL-based building extraction methods rely on an encoder–decoder architecture. The encoder encodes the original input image into low-resolution feature maps, while the decoder recovers pixel predictions from them. Typically, past classical methods [17] use skip connections to directly concatenate features from corresponding stages of the encoder and decoder along the channel dimension. However, semantic differences in the feature maps of different stages can lead to spatial dimension mismatches, thus limiting the accuracy of the constructed extraction. For the above reasons, some recent approaches are trying to combine the local contextual features provided by convolutional neural networks (CNNs) with the global contextual features provided by Transformers [18] to improve the network performance [19]. However, applications of [vision Transformer (ViT)] [20], [21], [22] tend to require large amounts of memory and computational cost. Therefore, finding an SA computation strategy that strikes a good balance between performance and computational cost is equally crucial for the building extraction task.

B. Underutilization of Cross-Modal Information

Recent studies have demonstrated the efficacy of incorporating multisource information in enhancing the robustness of building extraction. With advancements in airborne light detection and ranging (LiDAR) technology, a normalized

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digital surface model (nDSM) has emerged as a resource for extracting complementary features [22], [23]. By providing elevation data for objects, nDSM facilitates the segmentation of tall structures, such as buildings. However, it is susceptible to noise points, which can compromise data quality, underscoring the significance of integrating diverse cross-modal data. Currently, fusion methods can be broadly categorized into three main groups. Some algorithms directly integrate data from different modalities into a single feature extraction network at the input stage. Others employ separate networks to extract features from each modality and only fuse them during the final prediction stage [24]. However, these approaches do not fully exploit the information available in multiple modalities. At present, advanced fusion networks predominantly adopt feature-level fusion, where cross-modal features are extracted through independent branches and fused at various scales for utilization at different stages of the network [5], [25], [26]. These methods enable more comprehensive utilization of cross-modal information but necessitate an effective fusion strategy to extract complementary information from the cross-modal data while mitigating noise impact.

C. Rough Upsampling Structure

One of the crucial functions of the decoder is to upsample semantically rich visual features from low resolution to the input resolution. While some methods employ a simple combination of bilinear interpolation and convolution modules as an upsampling module [19], [29], this structure tends to be too coarse, resulting in the loss of fine-grained information during feature restoration. Especially, when it comes to the fine structure and textural details of an image, the simple combination approach struggles to capture the complex patterns and variations in the original image. Therefore, a rational and fine-grained upsampling module is an integral part of the building extraction network.

To tackle these challenging issues, we propose an attention-based cross-modal fusion network (ACMFNet) for extracting buildings in HSR remote sensing images. We employ a conventional encoder–decoder structure, where two identical backbone branches are utilized in the encoder to extract features from multispectral (MS) data and nDSM data. First, we use the global and local feature refinement module (GL-FRM) to establish a context dependency at multiple scales and levels through large-scale convolutions and attention computations. It combines the deep semantic feature maps of the decoder with the shallow fine-grained feature maps of the encoder, effectively solving the problem of spatial dimension mismatch. Second, in order to better extract valid information from multiple modalities, we use cross-modal complementary feature fusion module (CFM) for integrating the complementarities between two modalities. In addition, we introduce a lightweight residual upsampling module (RUM) to recover the resolution of these feature maps. The main contributions of this research can be summarized as follows.

1) We propose GL-FRM to enhance features and establish contextual dependencies at multiple scales and levels, mitigating the spatial discrepancy among multilevel features.
2) We introduce CFM to fuse MS data and nDSM data by weighting each modality and removing irrelevant parts.
3) A lightweight RUM is employed to restore the resolution of feature mappings.
4) The proposed ACMFNet achieves comparable results with state-of-the-art (SOTA) algorithms on two challenging benchmarks.

The remaining sections of this article are organized as follows. We first provide a brief introduction of related work in Section II. Then, in Section III, we present our proposed ACMFNet in detail. Furthermore, we report and analyze comprehensive experimental results in Section IV. Finally, conclusions and suggestions are given in Section V.

II. Related Work

In this section, we will review the recent literature from two perspectives: 1) representative works in MS building extraction and 2) representative works in cross-modal semantic segmentation.

A. MS Building Extraction

Compared with middle-/low-resolution remote sensing images, HSR remote sensing images offer more intricate details regarding land objects, thereby presenting challenges in building extraction tasks by intensifying intraclass variance and diminishing interclass variance [30]. Over the past two decades, numerous methodologies have been developed for building extraction from HSR remote sensing images. Early algorithms primarily relied on the physical principles of buildings [31], establishing criteria that best represent the local appearance of building roofs based on spatial or spectral features. Ngo et al. [32] employed geometric features to extract building regions, while Guo and Du [33] differentiated buildings from background through disparities in spectral reflectance. In addition, certain algorithms incorporated spatial constraints between buildings and their surrounding environment [34], such as shadows cast by buildings on the ground [35], to facilitate building extraction. Although these methods performed well for buildings governed by similar physical rules, they were unable to handle complex correlations between buildings and the background due to their limited universality and robustness.

With the rapid advancement of DL, CNN has emerged as the predominant approach for building extraction in remote sensing imagery in recent years. Mnih [35] was the first to introduce CNN for patch-based extraction of building roofs from high-resolution aerial images. Saito et al. [36] designed a neural network comprising three convolutional layers and two fully connected layers. Zuo et al. [16] incorporated multiscale features into their FCN-based model. Khalel and El-Saban [37] refined the output using U-Net with multiple stages, achieving relatively accurate building extraction. However, their models were limited by inadequate representation of global information. Consequently, attention mechanisms have been introduced in some studies to address this limitation. Guo et al. [38] proposed a multiloss neural network that enhances sensitivity and suppresses background interference

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through attention modules focused on irrelevant feature areas. Tian et al. [39] integrated a convolutional attention module into skip connection to extract salient multiscale features effectively. Das and Chand [40] employed various attention modules to improve overall feature representation. Nevertheless, these methods still heavily rely on convolutional operations without liberating the network from its CNN structure, thereby retaining certain limitations in modeling global information.

Methods based on ViT have brought great progress and development to semantic segmentation [41], [42], [43], and numerous researchers have incorporated ViT into building extraction tasks. Chen et al. [44] proposed a sparse token Transformer to effectively capture global dependencies among tokens in both spatial and channel dimensions. Wang et al. [45] introduced the Swin Transformer [46] as an encoder and devised a novel dense connection feature aggregation module for resolution restoration and segmentation map generation. Li et al. [29] designed an efficient linear complexity kernel attention mechanism to alleviate the computational requirements of attention operations. Some approaches combine CNN with Transformer to leverage their respective strengths in extracting global and local information. He et al. [46] and Wang et al. [19], respectively, proposed encoders with dual-path structures by parallelizing the Swin Transformer with CNN, enabling rich spatial details encoding through spatial context paths while capturing global dependencies via global context paths, thus achieving better results. Zhang et al. [47] designed an efficient dual spatial attention Transformer (DSAFormer) to solve the defects of standard ViT, which has a dual attention structure to complement each other.

B. Cross-Modal Semantic Segmentation

Single-modal semantic segmentation methods primarily emphasize the fusion of low-level spatial details and high-level semantic information. In contrast, cross-modal models integrate multiple dimensions of information, leveraging the complementarity across diverse data sources to extract more comprehensive features.

1) RGB-Depth (RGB-D) Semantic Segmentation: Successful RGB-D (RGB and depth) semantic segmentation heavily relies on the effective integration of cross-modal features. Most methods carry out feature fusion in the encoder. Wang et al. [48] proposed a feature transformation network to identify common features between cross-modal data and enhance the representation of shared information. Hazirbas et al. [49] introduced FuseNet, which integrates depth features at different levels into the RGB encoder as the network deepens. Jiang et al. [50] further extended FuseNet by incorporating a top-down path for fusing multilevel features. Hu et al. [51] proposed a three-branch network architecture and employed an attention complementary module to extract weighted features from both RGB and depth branches simultaneously. Chen et al. [52] introduced a unified and efficient cross-modal guidance encoder, recalibrated RGB feature responses, and accurately extracted depth information through multiple stages while alternately aggregating both modalities. Seichter et al. [53] presented efficient scene analysis network (ESANet) based on an improved ResNet encoder for efficient RGB-D semantic segmentation method that achieves higher accuracy with lower computational cost. Yue et al. [54] proposed a two-stage decoder that effectively integrates high-level and low-level features while suppressing low-order detail noise from shallow layers of the encoder. Zhang et al. [55] presented cross-modal fusion framework for RGB-X (CMX), which is a unified fusion framework utilizing cross-modal feature calibration module to calibrate bimodal characteristics and extensively exchange remote contexts via feature fusion modules.

2) MS and nDSM Building Extraction: The integration of nDSM data with MS data aims to address the information deficiencies associated with traditional single-modal processing. However, due to the high acquisition cost of nDSM data, there is limited research on semantic segmentation methods based on nDSM data sources. Audebert et al. [24] proposed an efficient multiscale method for semantic segmentation of remote sensing images, exploring early and late fusion strategies for nDSM and MS data. Peng et al. [22] introduced a dual-path dense convolution multimodal network that combines dense connections and FCN, extracting features from both MS and nDSM data before fusing them together. Nevertheless, their approach merely employed elementwise addition to fuse the two modalities' features, failing to fully exploit the rich information provided by nDSM data. Huang et al. [25], on the other hand, utilized an improved residual learning network as an encoder to learn multilevel features from fused data while introducing gated feature annotation units to reduce unnecessary feature transmission. Zhang et al. [26] adaptively selected and combined complementary features from each modality through an attention-aware multimodal fusion block in order to effectively learn and utilize multimodal information. Zhou et al. [56] proposed a gate fusion network, which incorporates a gate fusion module aimed at eliminating redundant features from both MS and nDSM data. Similarly, Hosseinpour et al. [5] used two independent encoders for extracting multilevel features from MS and nDSM data while achieving effective fusion through a gate fusion module. Hong et al. [27] introduced SpectralGPT, the first foundation model designed specifically for spectral remote sensing data, which also demonstrates significant potential in advancing cross-modal semantic segmentation tasks [28].

III. Method

In this section, we describe the proposed ACMFNet framework in detail. Our ACMFNet follows the encoder–decoder structure, as visualized in Fig. 1. GL-FRM, CFM, and RUM make up our proposed method's three main parts. First, the overall framework employed in this work will be introduced in Section III-A. Then, we will go into the details of the proposed modules in Sections III-B–III-D.

A. Framework Overview

The architecture of ACMFNet, as illustrated in Fig. 1, corresponds to a depth-supervised encoder–decoder network. In the encoder, we have incorporated two separate branches for extracting features from MS data and nDSM data. For these

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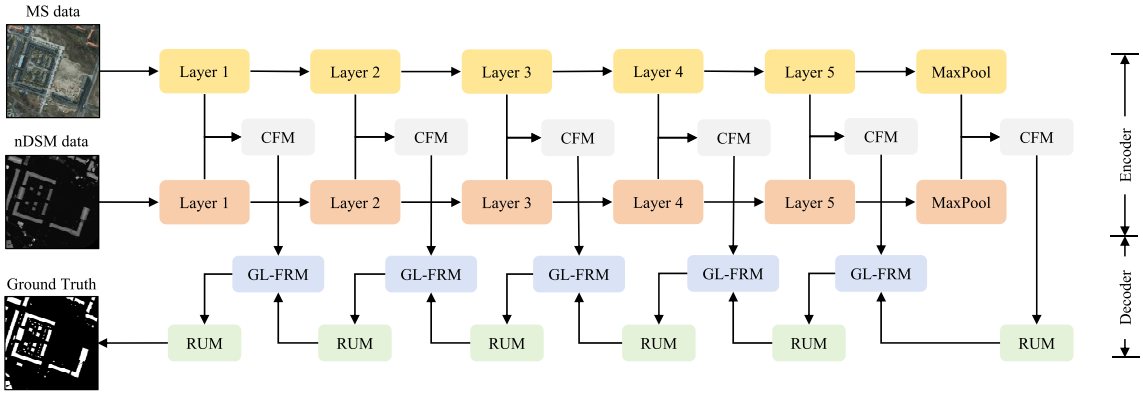


Fig. 1. Architecture of our proposed ACMFNet. ACMFNet contains three important modules: CFM, GL-FRM, and RUM.

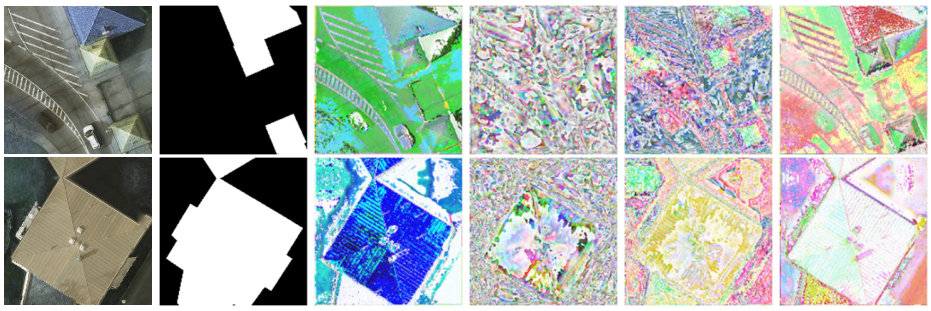


Fig. 2. (a) Original image. (b) Ground truth. (c) Shallow fine-grained feature from encoder. (d) Deep semantic feature from decoder. (e) Fusion by CAT. (f) Fusion by GL-FRM.

branches, ResNet-34 [57] has been chosen as the backbone due to its moderate depth and residual structure, which are compatible with real-time operations owing to their smaller operations. To suit the semantic segmentation task requirements, all fully connected layers of ResNet-34 have been removed. Except for the first convolution module with a single channel on the nDSM data branch, both branches share an identical network configuration comprising six module stages that include five convolutional layers and one max-pooling layer for extracting multiscale features. After each feature extraction stage, CFM is employed to extract complementary parts between the two modalities for feature recombination. In the decoder, GL-FRM establishes a context dependency at multiple scales and levels through large-scale convolutions and attention computations by combining deep semantic feature maps from decoder with shallow fine-grained feature maps from encoder, which effectively addresses spatial dimension mismatch issues between them. Finally, lightweight RUM is utilized to restore resolution in these feature maps.

B. GL-FRM

During the process of recovering pixel predictions from low-resolution feature maps in the decoder, incorporating feature maps from corresponding scales in the encoder to deep semantic information can effectively provide multiscale and multilevel contextual information, thereby enhancing segmentation precision. It is a common practice to directly concatenate features along the channel dimension. However, as depicted in Fig. 2(c) and (d), these two types of features exhibit semantic disparities that result in feature misalignment as the network goes deeper. Hence, explicit and dynamic establishment of position correspondence between feature maps becomes necessary [59]. To address this issue, we propose GL-FRM that leverages extensive convolutional operations and attention calculations to establish multiscale and multilevel contextual dependencies, thus alleviating mismatches between deep semantic information and shallow fine-grained spatial details.

The deep semantic feature X_d from the decoder and the shallow fine-grained feature X_e from the encoder are initially concatenated in the channel dimension to obtain X, as illustrated in Fig. 3. To refine local features, a sparsely connected architecture is employed to stack various types of convolution kernels (3×3, 5×5, 7×7, and MaxPool) for extracting densely refined data X_l [60]

$$\begin{aligned} X &= \mathrm{Cat}(X_e, X_d) \\ X_l &= \mathrm{Cat}(\mathrm{Conv}_{3\times3}(X), \mathrm{Conv}_{5\times5}(X), \mathrm{Conv}_{7\times7}(X), \\ &\qquad \mathrm{MaxPool}(X)) \end{aligned} \quad (1)$$

where Cat denotes the channelwise concatenation (CAT) and $\mathrm{Conv}_{a\times a}$ denotes a convolutional layer with a convolutional kernel size of $a \times a$.

For global feature refinement, we aim to leverage the powerful coding capability of Transformer for conducting feature refinement and exploring correlations between channels. In traditional ViT, a multihead SA mechanism is employed where the input image sequence X is multiplied by three weight matrices W_q, W_k, and W_v to derive query (Q), key (K), and value (V). By utilizing dot product calculations, the query tokens are individually matched separately with the key tokens,

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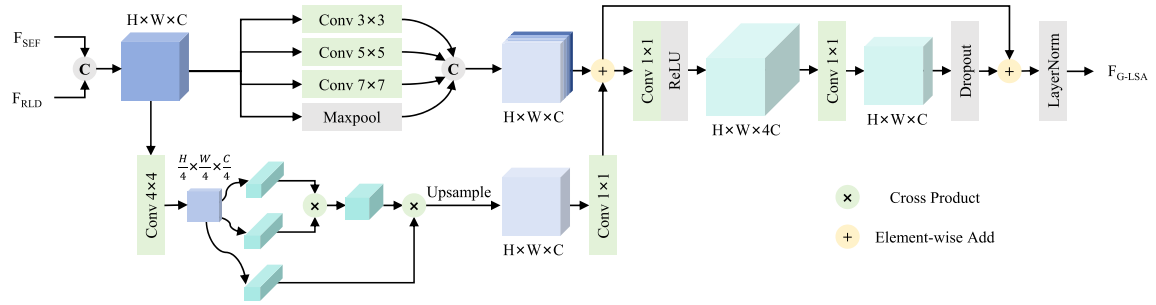


Fig. 3. GL-FRM.

resulting in an attention map that reflects token similarity. The value component represents the important features extracted from each token, which are then multiplied with the attention map. Through such computations performed on all tokens, SA facilitates global information exchange. The process can be illustrated as follows:

$$Q, K, V = XW_q, XW_k, XW_v$$
$$\mathrm{SA}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{2}$$

where Softmax represents the Softmax function and SA represents the SA calculation. However, this mechanism also incurs a substantial computational cost that scales quadratically with the number of tokens. Wang et al. [21] employ a strategy of reducing the sequence length during SA calculation to reduce the computational cost. However, in order to maintain the output size, the query is still processed at full resolution, resulting in significant computational overhead. Considering the localized nature of buildings in HSR images, dense pixel-wise SA is unnecessary for establishing global dependencies. To reduce computational cost, we simultaneously decrease the length of the Q, K, and V sequences to establish coarse global dependencies, further reducing the computational cost. We employ nonoverlapping 4×4 convolutions to merge adjacent pixel tokens, thereby reducing burden during SA calculations. Subsequently, we upsample the feature maps to match the input size, denoted as X_g. The refinement features X_l and X_g are concatenated and fed into a multilayer perceptron (MLP) for further information fusion. The resulting output is denoted as Z

$$X' = \mathrm{Conv}_{4\times4}(X)$$
$$Q, K, V = X'W_q, X'W_k, X'W_v$$
$$X_g = \mathrm{Upsample}(\mathrm{SA}(Q, K, V))$$
$$Z = \mathrm{MLP}\left(X_g + X_l\right) \tag{3}$$

where Upsample denotes bilinear interpolation upsampling and MLP denotes MLP calculation.

C. CFM

In the task of building extraction in cross-modal HSR remote sensing imagery, it is crucial to determine the optimal utilization of MS and nDSM information. MS encompasses a wealth of spectral information pertaining to objects and scenes, while nDSM provides additional contour and positional details. The inherent disparities between these two

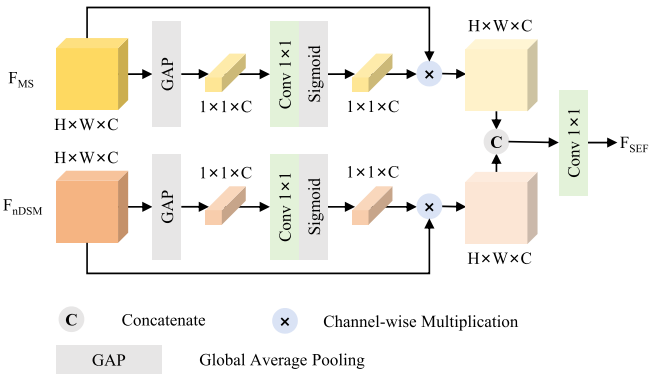


Fig. 4. Cross-modal complementary feature fusion module.

data modalities may lead to increased noise or redundancy when integrating features. To effectively integrate MS and nDSM information, we proposed CFM, enabling the network to focus on acquiring more complementary informational features from both MS and nDSM sources. As illustrated in Fig. 4, we implemented a channel attention approach based on squeeze-and-excitation [58] mechanism that learns to accentuate valuable channels using global information while suppressing less informative ones.

In a multibranch architecture, we have MS input feature maps X_M and nDSM input feature maps X_D. For both branches, we use global average pooling (GAP) as the channel descriptor for channel attention in both branches, followed by a 1×1 convolutional block to explore interchannel correlations. The resulting convolutional output is activated using the sigmoid function to restrict the weight vector values between 0 and 1. Subsequently, these weight vectors are multiplied with input feature maps from both branches to obtain X'_M and X'_D. Due to an increase in dimensionality caused by concatenating feature maps, there is an excessive computational burden and inclusion of redundant information. To address this issue, we utilize a 1×1 convolutional layer to reduce the feature dimensionality to 32 for eliminating redundancy. The formulaic representation of the CFM module is shown as follows:

$$X'_M = X_M \otimes \sigma(\mathrm{Conv}_{1\times1}(\mathrm{AvgPool}(X_M)))$$
$$X'_D = X_D \otimes \sigma(\mathrm{Conv}_{1\times1}(\mathrm{AvgPool}(X_D)))$$
$$Z = \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(X'_M, X'_D\right)\right) \tag{4}$$

where Z denotes the final output feature representation, AvgPool denotes GAP, and σ denotes the sigmoid function.

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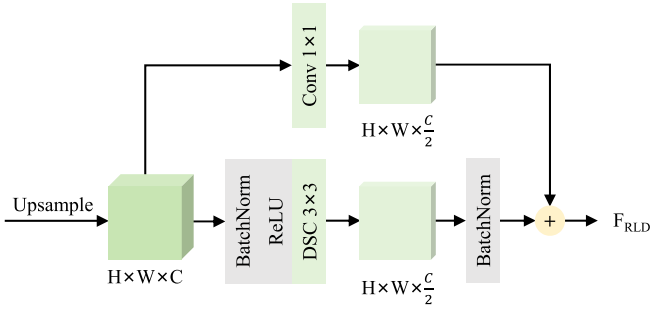


Fig. 5. RUM.

D. RUM

An excessively simple upsampling structure will lead to coarse extractions. To address this limitation, we propose an enhanced lightweight RUM, as illustrated in Fig. 5. The input denoted as X is first upsampled using bilinear interpolation to match the resolution of lateral features, yielding X_{up}. In the residual branch, we explore channel correlations and reduce feature dimensions through a 1×1 convolution block denoted as X_{res}. Simultaneously, in the main branch, X_{up} is computed by applying BatchNorm, activation function, and convolution layers to obtain feature X_{conv}, which is then combined with X_{res} to produce output feature Z. This residual structure effectively enhances and preserves more original features. Moreover, within RUM architecture, standard convolution modules are replaced with depthwise separable convolution (DSC) [61] modules for parameter reduction purposes without compromising performance quality or sacrificing network efficiency in the decoder component. Consequently, the resolution of feature maps is doubled, while the number of channels is halved. The overall process description for RUM can be summarized as follows:

$$\begin{aligned} X_{\text{up}} &= \text{Upsample}(X) \\ X_{\text{res}} &= \text{Conv}_{1\times1}\left(X_{\text{up}}\right) \\ X_{\text{conv}} &= \text{DeepConv}\left(X_{\text{up}}\right) \\ Z &= X_{\text{res}} + X_{\text{conv}} \end{aligned} \tag{5}$$

where DeepConv consists of a BatchNorm layer, rectified linear unit (ReLU) activation function, and DSC layer. The decoder employs six RUMs to progressively upsample the low-resolution feature maps, aligning them with the resolution of the input.

IV. Experiments and Analysis

This section is divided into four parts. First, we provide a detailed description of the two benchmark datasets. Second, the whole implementation process is comprehensively understood through training settings, loss function, and evaluation metrics. After that, an ablation study is conducted to verify the effects of each proposed module. Finally, we quantitatively and qualitatively compare the proposed ACMFNet with 16 competitive methods in the form of tables and pictures, respectively.

A. Datasets

In this article, two benchmark datasets, namely, Vaihingen and Potsdam, were used to verify the effectiveness of the proposed ACMFNet, both of which encompass urban scenes and were provided by Commission II/4 of the ISPRS.

1) Vaihingen Dataset: Vaihingen is a relatively small village characterized by numerous small multistory buildings and standalone structures. Vaihingen [62] consists of 33 true orthophoto (TOP) images captured by advanced airborne sensors, along with their corresponding DSM data, covering an area of 1.38 km^2 in the town of Vaihingen, Germany. Each image patch has varying dimensions, approximately 2000 $\times$ 2500 pixels, with a ground sampling distance (GSD) of around 9 cm. These patches contain only three bands [Vaihingen-infrared-red-green (IRRG)], namely, infrared (IR), red (R), and green (G). The original ground truth encompasses six primary land cover classes: impervious surface, buildings, low vegetation, trees, cars, and clutter. For the purpose of building extraction in this study, foreground and background classes are utilized. According to the data provider, a total of 16 patches were selected for training, while the remaining patches were allocated for testing.

2) Potsdam Dataset: Potsdam is a quintessential historical city characterized by expansive architectural areas, narrow streets, and a densely populated settlement structure. Similar to Vaihingen, the dataset for Potsdam [62] comprises 38 TOP images accompanied by corresponding DSM patches, encompassing an area of 3.42 km^2 within the city of Potsdam in Germany. Each patch measures 6000 $\times$ 6000 pixels and has a GSD of 5 cm. These patches are available in two modes, the red (R), green (G), and blue (B) bands (Potsdam-RGB), and the near infrared (NIR), red (R), and green (G) bands (Potsdam-IRRG). The classes in the ground-truth data align with those used in Vaihingen. Specifically, the training set consists of 24 RGB patches, while the remaining patches are allocated for testing purposes using RGB imagery.

B. Implementation Details

1) Loss Function: In reference to BuildFormer [19], we trained ACMFNet with a joint loss function using the combined incorporation of boundary supervision technology. The joint loss function can be defined as follows:

$$L = L_{\text{ce}}\left(Y, \hat{Y}\right) + L_{\text{dice}}\left(Y, \hat{Y}\right) + L_{\text{bce}}\left(\mathcal{L}(Y), \mathcal{L}\left(\hat{Y}\right)\right) \tag{6}$$

where Y and $\hat{Y}$ represent the predicted label and true label, L_{ce} is the cross-entropy loss, L_{dice} is the dice loss, $\mathcal{L}$ represents the Laplacian convolution used to extract building boundaries from predicted labels and true labels, and L_{bce} represents the binary cross-entropy loss applied to the building boundaries.

2) Evaluation Metrics: To quantitatively evaluate the performance of our method and other methods involved in the comparison, we use intersection over union (IoU), F1 score, precision, and recall, which are widely used in building extraction.

For each category, IoU is defined as the ratio of intersection and union of the predicted value and the true value

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}. \tag{7}$$

F1 score is a classical criterion for binary classification between interest targets and nontargets, which is equal to the

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TABLE I
IoU AND THE CORRESPONDING TIME REQUIRED TO TRAIN EACH EPOCH FOR VARIOUS STRIDES ON POTSDAM DATASET

Stride	The amount of training data	IoU (%)	Training time/epoch (min)
512×512	3456	90.76	1.76
384×384	6144	92.59	3.15
256×256	12696	93.13	6.47
128×128	50784	93.15	20.88

harmonic mean of precision and recall

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

where TP, FP, and FN represent true positive, false positive, and false negative, respectively.

3) Training Settings: The original images from Potsdam and Vaihingen datasets were cropped into the dimensions of 512 × 512 using a sliding window. By using the Potsdam dataset as an example, we presented IoU and the corresponding time required to train each epoch for various strides in Table I. It was observed that these two metrics achieved optimal performance at 256 × 256 strides; therefore, we have chosen to set the stride length accordingly. To effectively expand the training data and mitigate overfitting, we randomly augmented the training set by applying horizontal and vertical flips as well as rotations of 90°, 180°, and 270°.

To ensure fairness, all experiments were conducted under identical conditions. All models used in the experiments were implemented using PyTorch 1.10.0 (CUDA 11.3) on an NVIDIA GTX 4090 GPU with 24-GB RAM. During the training phase, we employed the AdamW optimizer and utilized a cosine strategy to adjust the learning rate accordingly, which facilitates the model in escaping local optima during training and converging toward a global optimal solution. Specifically, we set the initial learning rate to 0.001, the initial restart epoch T_0 to 15, and the learning rate recovery speed T_{mult} to 2. A batch size of 8 was used, and a maximum of 105 epochs were executed. The training loss of ACMFNet on two datasets is illustrated in Fig. 6.

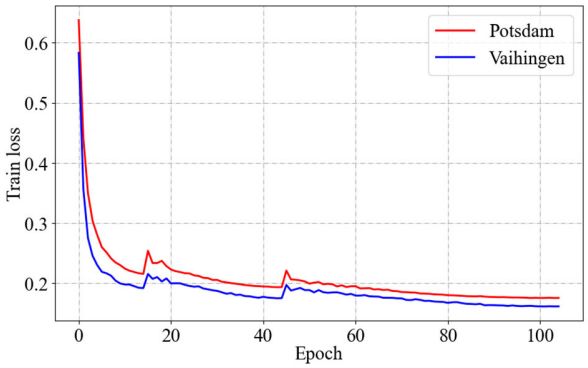


Fig. 6. Training loss of ACMFNet on Vaihingen and Potsdam datasets.

C. Ablation Study

To validate the effectiveness of the proposed ACMFNet, extensive ablation experiments are presented in this section.

We conduct ablation experiments on Vaihingen and Potsdam datasets to evaluate the contribution of each proposed module to our ACMFNet, and the results are shown in Table II, which demonstrates the IoU and F1 score of different variations. We note that the baseline uses elementwise summation (SUM) to combine data from both modalities. On Vaihingen, by solely introducing CMF, IoU and F1 exhibit a growth of 0.48% and 0.23%, respectively. Upon simultaneous introduction of CFM and GL-FRM, the result is improved by 0.78% IoU and 0.38% F1. Further enhancement is observed with the concurrent introduction of CFM, GL-FRM, and RUM, resulting in an increase of 1.26% for IoU and 0.61% for F1 score. On Potsdam, the incorporation of these three modules also demonstrates a similar positive trend in terms of IoU and F1 score improvement as observed in Vaihingen. The results in Table II verify that each component of ACMFNet helps boosts building extraction performance.

1) Effect of Multilevel Feature Fusion and GL-FRM: We conducted a study on the efficacy of multilevel feature fusion and GL-FRM, and the results are presented in Table III. In ACMFNet, common multilevel feature fusion is incorporated into the encoder–decoder architecture, where the fused features from CFM in the encoder are concatenated with the high-level features in the decoder through skip connections. To explore their effectiveness, we eliminated skip connections from ACMFNet. The findings revealed a decrease in IoU by 1.39% and 0.84% on Vaihingen and Potsdam, respectively. Furthermore, we investigated the effectiveness of GL-FRM in multilevel feature fusion by replacing it with channelwise CAT, a commonly used method, resulting in an experiment that demonstrated significant improvements in building extraction results, an increase of 0.47% IoU on Vaihingen dataset and 0.18% on Potsdam dataset. Furthermore, relying solely on the global feature or the local feature results in a decline in performance.

As depicted in Fig. 7, relying solely on the deep semantic information of the decoder for feature resolution recovery results in a significant loss of spatial details. To address this limitation, introducing fine-grained features at the corresponding scale of the encoder can offer multiscale and multilevel information, leading to clearer contours and finer segmentation effects. However, directly concatenating fea-

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TABLE II

ABLATION EXPERIMENTS OF ACMFNET. THE BEST PERFORMANCE IS DISPLAYED IN BOLD. THE EFFECTIVENESS OF THREE PROPOSED MODULES IS VERIFIED BY THE PERFORMANCE CHANGE RELATIVE TO THE BASELINE. THE PROPOSED ACMFNET COMBINES THEIR BENEFITS TO MAXIMIZE THE BUILDING EXTRACTION PERFORMANCE

baseline	CFM	GL-FRM	RUM	Vaihingen (%)		Potsdam (%)	
				IoU	F1	IoU	F1
√				88.23	93.85	92.43	96.07
	√			88.71	94.08	92.58	96.11
	√	√		89.01	94.23	92.79	96.24
	√	√	√	**89.76**	**94.60**	**93.13**	**96.44**

TABLE III

EVALUATION METRICS SCORE OF ABLATION EXPERIMENT FOR MULTILEVEL FEATURE FUSION AND GL-FRM

Methods	Vaihingen (%)		Potsdam (%)	
	IoU	F1	IoU	F1
ACMFNet without multi-level feature fusion	88.37	93.82	92.29	95.99
Replace GL-FRM with CAT	89.29	94.34	92.95	96.35
GL-FRM without global stream	89.40	94.41	93.01	96.37
GL-FRM without local stream	89.61	94.52	93.07	96.42
ACMFNet with GL-FRM	89.76	94.60	93.13	96.44

TABLE IV

EVALUATION METRICS SCORE OF ABLATION EXPERIMENT FOR CROSS-MODAL FUSION AND CFM

Methods	Vaihingen (%)		Potsdam (%)	
	IoU	F1	IoU	F1
ACMFNet without nDSM	88.41	93.85	90.42	94.97
Replace CFM with SUM	89.40	94.40	92.89	96.32
Replace CFM with CAT	89.45	94.43	93.04	96.39
ACMFNet with CFM	89.76	94.60	93.13	96.44

TABLE V

EVALUATION METRICS SCORE OF ABLATION EXPERIMENT FOR RUM

Methods	Vaihingen (%)		Potsdam (%)	
	IoU	F1	IoU	F1
RUM without residual	89.16	94.27	92.94	96.26
Replace DSC with Conv	89.10	94.24	92.85	96.29
ACMFNet with GL-FRM	89.76	94.60	93.13	96.44

ture maps using CAT introduces semantic differences and spatial dimension mismatches [see Fig. 2(e)]. In contrast, GL-FRM effectively mitigates this issue by establishing context dependencies through extensive convolution and attention computations across multiple scales and levels. This approach supplements extracted features with abundant global information while matching and integrating information from both feature maps. Consequently, it refines building-related details further, excludes background noise effectively, and enhances the accuracy of building extraction [see Fig. 2(f)].

2) Effect of Cross-Modal Fusion and CFM: We investigated the efficacy of cross-modal fusion and CFM, and the corresponding results are presented in Table IV. To validate the effectiveness of integrating nDSM into building extraction tasks, we completely eliminated the nDSM stream from ACMFNet, consequently removing CFM as well.

The findings demonstrated a significant enhancement in building extraction results when utilizing nDSM, with an increase in IoU by 1.35% and 2.71% for Vaihingen and Potsdam datasets, respectively. Subsequently, we replaced CFM with two fusion techniques, elementwise SUM and CAT, aiming to explore the efficacy of CFM in cross-modal fusion. The outcomes obtained from these two fusion methods exhibited similarity, with CAT slightly surpassing SUM. However, both fell short compared with the results achieved using CFM.

As depicted in Fig. 8, when dealing with roof structures that exhibit similar texture features to the ground and shadows, relying solely on MS can result in missed detections and false positives. However, incorporating the nDSM stream significantly mitigates this issue. Eliminating CFM compromises the integrity of building extraction outcomes. On the one hand, MS data are prone to noise, such as light interference, and there exists a significant presence of shadows caused by surrounding buildings. On the other hand, nDSM contains a certain amount of noise points. Moreover, it poses a challenge to differentiate buildings from objects of similar height. Inherent disparities exist between these two modalities. Thus, simply performing SUM or CAT operations would undermine the accuracy of building extraction results. Nevertheless, employing CFM enables the extraction of complementary information from both modalities and, thereby, enhances the precision of building extraction outcomes.

3) Effect of RUM: We conducted a study to evaluate the efficacy of the proposed RUM, and the experimental results are presented in Fig. 9 and Table V. The utilization of residual structures led to an improvement in IoU by 0.60% and 0.19% on Vaihingen and Potsdam datasets, respectively, owing to the network's ability to preserve more original semantic information during the upsampling process. Furthermore, substituting regular 3×3 convolutions with DSC resulted in an IoU increase of 0.66% and 0.28% while reducing parameter count by approximately 0.1 M. By employing RUM, we facilitated the transfer of richer semantic information related to building objects from deep convolution blocks to lower layers during upsampling, thereby achieving superior performance in building extraction tasks while maintaining a lightweight network structure.

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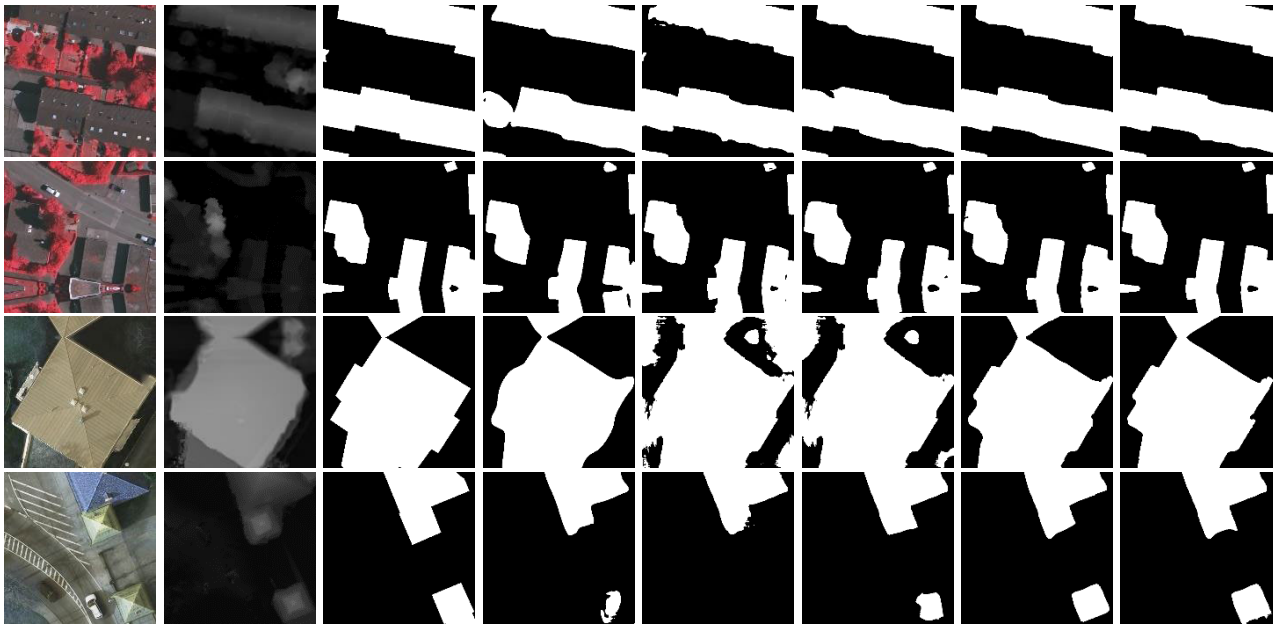


Fig. 7. Visualization results of ablation experiments on multilevel feature fusion and GL-FRM. (a) Original image. (b) nDSM. (c) Ground truth. (d) ACMFNet without multilevel feature fusion. (e) Replace GL-FRM with CAT. (f) GL-FRM without global stream. (g) GL-FRM without local stream. (h) ACMFNet (proposed).

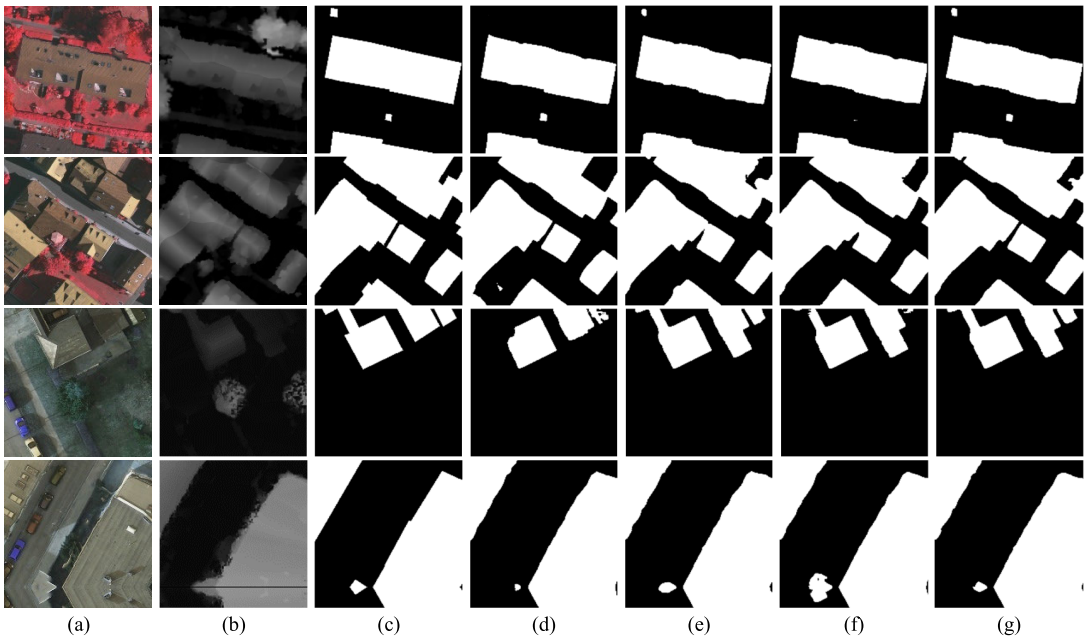


Fig. 8. Visualization results of ablation experiments on cross-modal fusion and CFM. (a) Original image. (b) nDSM. (c) Ground truth. (d) ACMFNet without nDSM. (e) Replace CFM with SUM. (f) Replace CFM with CAT. (g) ACMFNet (proposed).

D. Comparisons With Other Methods

We compared ACMFNet with 16 DL-based semantic segmentation methods, including U-Net [14], FCN [13], DeepLabv3+ [63], dense dilated convolutions merging network (DDCM-Net) [64], multi-attention-network (MA-Net) [29], DC-Swin [45], BuildFormer [19], dual spatial attention Transformer net (DSATNet) [47], V-FuseNet [24], cross-modal gated fusion network (CMGFNet) [5], FuseNet [49], RedNet [50], attention complementary network (ACNet) [51], SA-Gate [52], ESANet [53], and CMX [55]. As the methods in [13], [14], [19], [29], [45], [63], and [64] are all based on single modal, we only used MS data. To ensure a fair comparison, the first four methods all chose ResNet-34 as backbone. Since the methods in [49], [50], [51], [52], [53], and [55] are all semantic segmentation methods for RGB-D scenes, we used nDSM instead of depth data. The building extraction results for all methods were obtained by running available codes.

1) Comparisons With Single-Modal Methods: Table VI shows the numerical results of single-modal semantic segmentation methods used for building extraction. Our proposed method achieved an IoU of 88.41% on Vaihingen, which is a 0.22% improvement over BuildFormer, and outperformed other algorithms in terms of precision and F1. In addition, our method achieved the best IoU, recall, and F1 scores

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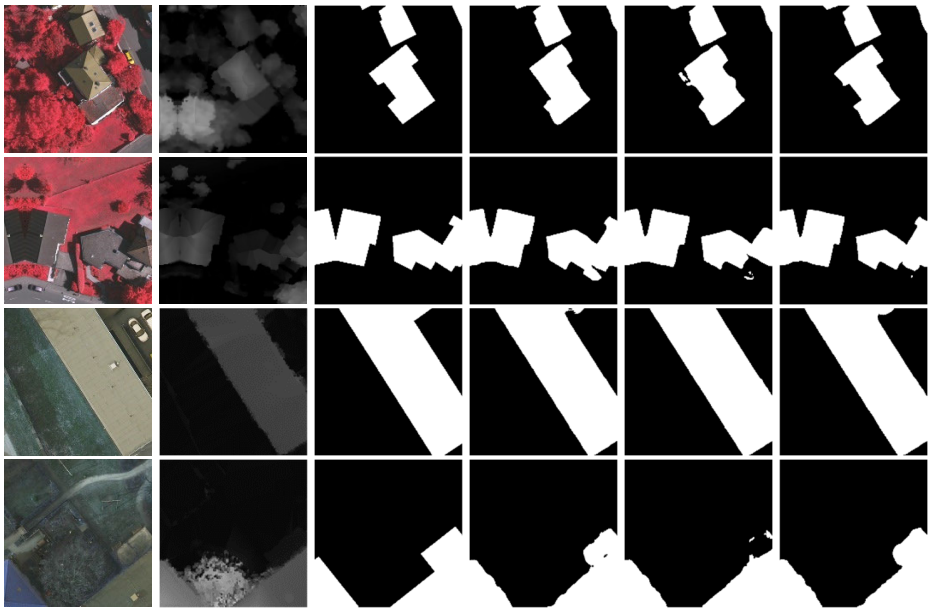


Fig. 9. Visualization results of ablation experiments on RUM. (a) Original image. (b) nDSM. (c) Ground truth. (d) RUM without residual. (e) Replace DSC with conventional convolution. (f) ACMFNet (proposed).

TABLE VI
COMPARISON OF DIFFERENT SINGLE-MODAL NETWORKS WITH THE PROPOSED METHOD

Methods	Vaihingen				Potsdam			
	IoU	Precision	Recall	F1	IoU	Precision	Recall	F1
U-Net	87.40	93.55	93.00	93.28	90.20	94.87	94.82	94.85
FCN8s	87.32	93.17	93.29	93.23	89.65	94.74	94.35	94.54
DeepLabv3+	87.56	93.37	93.37	93.37	90.01	94.74	94.74	94.74
MA-Net	87.12	93.52	92.72	93.11	89.96	**95.27**	94.17	94.72
DC-Swin	81.90	90.02	90.08	90.05	84.81	92.98	90.61	91.78
ST-UNet	83.81	91.26	91.12	91.19	87.40	93.34	93.21	93.27
BuildFormer	88.19	93.74	**93.71**	93.73	90.20	95.05	94.65	94.85
DSATNet	88.08	93.65	93.68	93.66	90.02	94.89	94.61	94.75
ACMFNet without nDSM	**88.41**	**94.11**	93.60	**93.85**	**90.42**	94.76	**95.17**	**94.97**

TABLE VII
COMPARISON OF DIFFERENT CROSS-MODAL NETWORKS WITH THE PROPOSED METHOD

Methods	Vaihingen				Potsdam			
	IoU	Precision	Recall	F1	IoU	Precision	Recall	F1
V-FuseNet	87.44	93.34	93.26	93.30	92.11	95.56	96.23	95.89
CMGFNet	88.81	94.52	93.63	94.07	92.61	96.11	96.22	96.16
FuseNet	87.82	93.46	93.57	93.51	92.32	95.72	96.30	96.01
RedNet	88.55	93.75	94.11	93.93	92.35	95.64	96.41	96.02
ACNet	88.51	94.22	93.60	93.31	92.63	**96.12**	96.24	96.18
SA-Gate	88.77	94.76	93.35	94.05	92.59	96.04	96.27	96.15
ESANet	88.83	94.45	93.68	94.08	92.43	96.04	96.10	96.07
CMX	89.01	94.14	94.23	94.19	92.59	95.80	96.51	95.15
ACMFNet	**89.76**	**94.86**	**94.35**	**94.60**	**93.13**	95.90	**96.98**	**96.44**

on Potsdam. To qualitatively analyze the building extraction performance of the proposed ACMFNet, we visualized its results alongside the results of the other three competitive methods as well as the ground truth. From Fig. 10, it is

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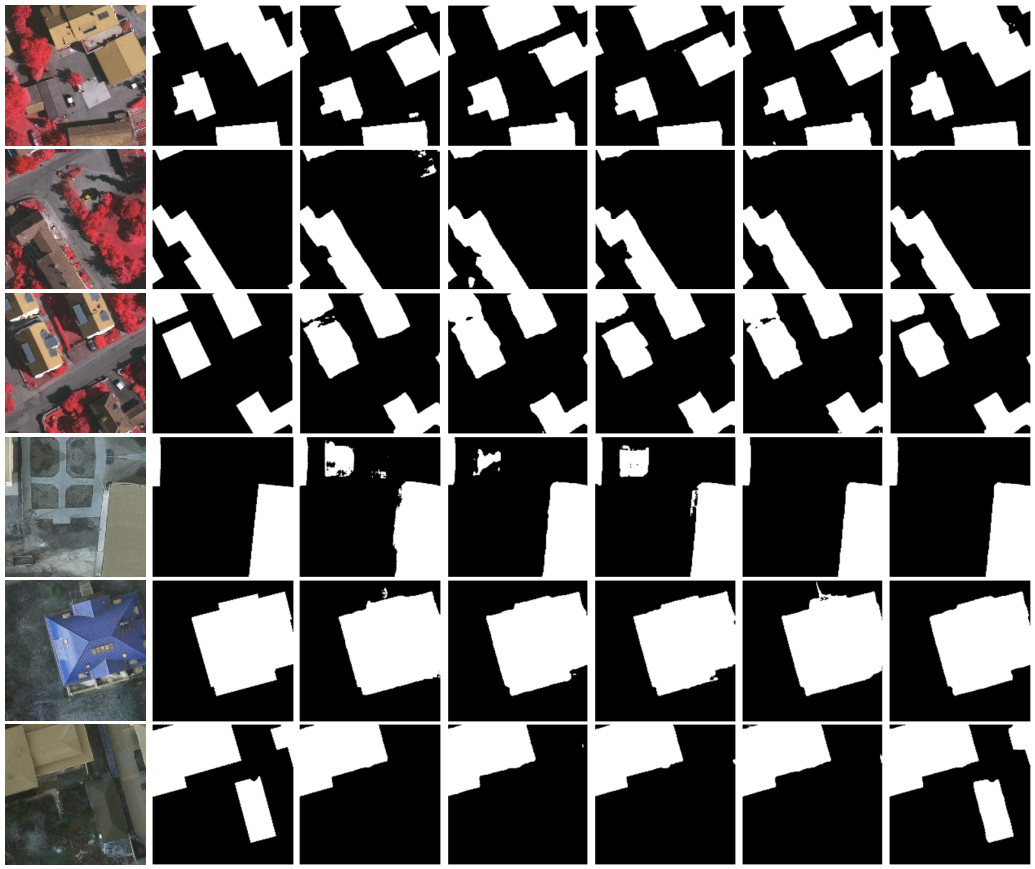


Fig. 10. Visualization results compared with single-modal methods. (a) Original image. (b) Ground truth. (c) DeepLabv3+. (d) MA-Net. (e) DSATNet. (f) BuildFormer. (g) ACMFNet without nDSM (proposed).

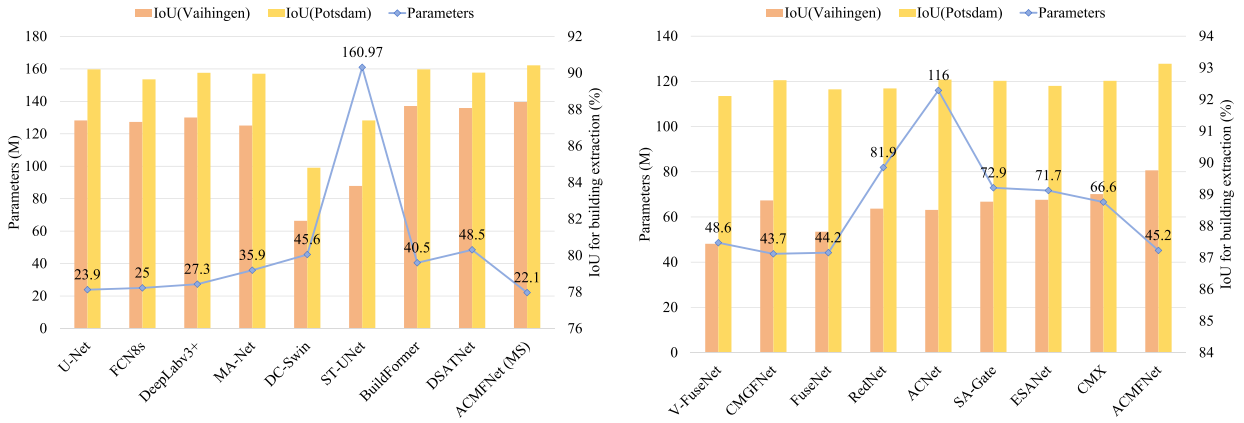


Fig. 11. Parameters and IoU of our ACMFNet with other methods. (a) Comparisons with single-modal methods. (b) Comparisons with cross-modal methods.

easy to find that compared with the other three methods, ACMFNet has a stronger ability to distinguish the buildings from the background, while the edges of the extracted buildings are clearer. DeepLabv3+ utilizes a unique spatial pyramid pooling module and decoder structure, resulting in superior performance compared with other CNN-based methods. U-Net integrates spatial information from lower level features through skip connection, yielding segmentation results slightly inferior to DeepLabv3+. In contrast, BuildFormer incorporates an additional global contextual pathway in the encoder to capture global dependencies, providing better global modeling capability. However, its overly simplistic decoder structure leads to rough resolution recovery. Our proposed method utilizes ResNet-34 as the encoder and incorporates GL-FRM in the decoder to match deep semantic features with shallow fine-grained features and establish contextual dependencies at the spatial level, which resulted in better accuracy in recognizing building pixels and maintaining building integrity. In addition, Fig. 11(a) shows the parameters for each method, with ST-U-shaped network (ST-UNet) having a parameter

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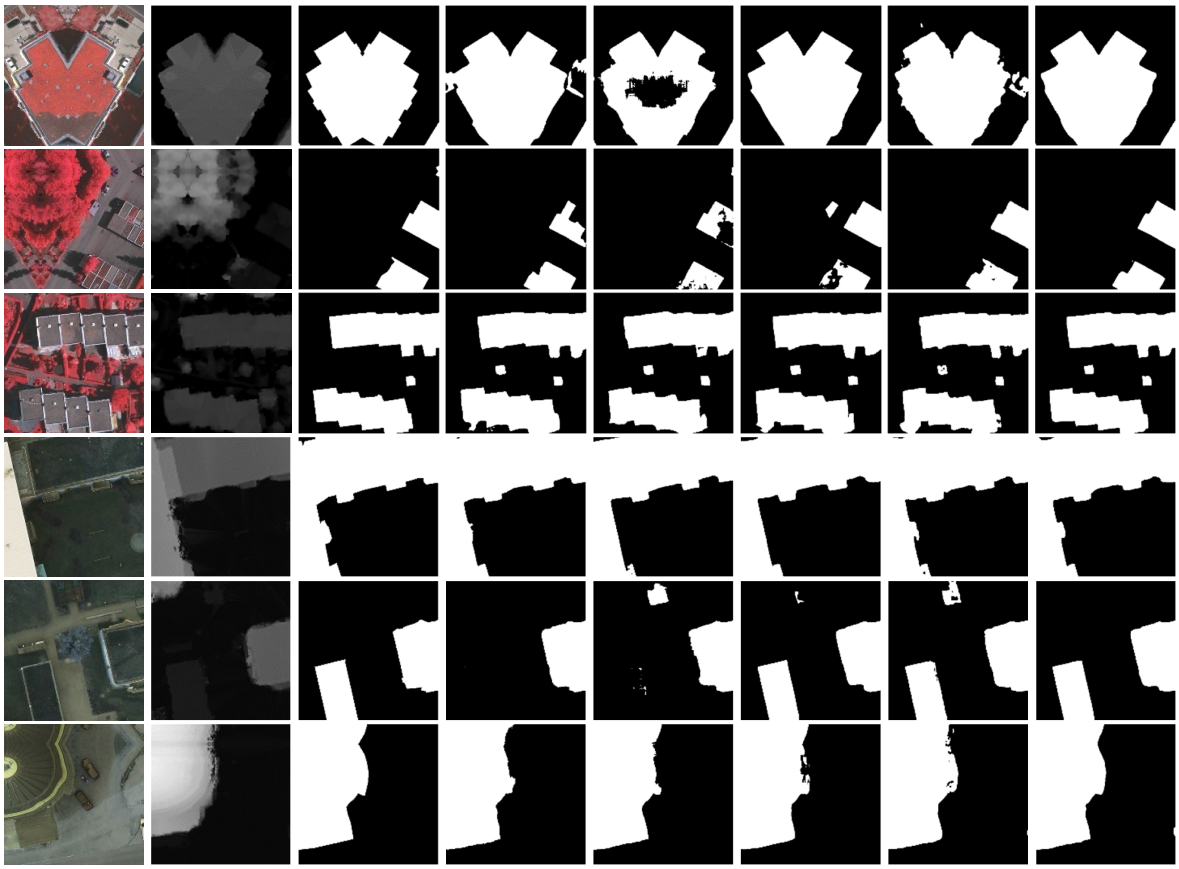


Fig. 12. Visualization results compared with cross-modal methods. (a) Original image. (b) nDSM. (c) Ground truth. (d) CMGFNet. (e) ACNet. (f) ESANet. (g) CMX. (h) ACMFNet (proposed).

count of 160.97 M, far exceeding other methods but yielding inferior results. This indicates that using a complex encoder structure does not necessarily lead to performance improvements and may instead increase model complexity, resulting in unnecessary computations. Our proposed method achieved optimal results while significantly reducing the parameter count because of dimension reduction of features at each stage of the encoder and the lightweight decoder module.

2) Comparisons With Cross-Modal Methods: Table VII presents the numerical results of cross-modal semantic segmentation methods employed for building extraction. Our proposed method achieved the highest IoU of 89.76%, demonstrating a notable improvement of 0.95% compared with CMGFNet (88.81%) on the Vaihingen dataset. Furthermore, our approach outperformed other algorithms in terms of precision, recall, and F1 scores. On the Potsdam dataset, our method attained an impressive IoU score of 93.13% and surpassed alternative algorithms in recall and F1 metrics as well. Particularly noteworthy is our algorithm's exceptional recall rate of 96.98%, surpassing other approaches by a significant margin of 0.76%, which indicates fewer instances where building pixels were missed during segmentation process. We selected four methods with relatively good performance to visualize the results in Fig. 12. In comparison with other cross-modal methods, our proposed ACMFNet demonstrates accurate segmentation of building areas across various challenging scenarios. When dealing with buildings heavily affected by shadows or tree height, our method excels at distinguishing nonbuilding areas while maintaining higher integrity in building extraction due to the fusion of complementary features in cross-modal data through CFM. Moreover, for buildings exhibiting irregular geometric structures, our approach achieves clearer and more complete boundary structures by leveraging comprehensive integration and utilization of high-level semantic features, low-level spatial details, and global contextual information via GL-FRM. In addition, as shown in Fig. 11(b), our method outperforms RGB-D-based semantic segmentation methods while significantly reducing parameter count. Compared with the SOTA method CMGFNet for cross-modal building extraction, our method introduces fewer additional parameters yet yields significant improvements in extraction results.

V. Conclusion

In this article, we propose ACMFNet, a novel building extraction method for cross-modal HSR remote sensing images that adopts an encoder–decoder structure. To alleviate spatial mismatch, we propose GL-FRM, which establishes multiscale and multilevel contextual dependency by combining large-scale convolution and attention computation. It integrates deep semantic feature maps from the decoder with shallow fine-grained feature maps from the encoder. Furthermore, CFM is employed to integrate the complementary parts between MS data and nDSM data. In addition,

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a lightweight RUM is used to recover the resolution of the feature maps. Extensive experimental results on Vaihingen dataset and Potsdam dataset show that the proposed ACMFNet achieves superior building extraction performance. However, limitations exist in building boundary extraction where segmentation results do not align perfectly with building shapes and boundaries appear less smooth. In future work, we will explore encoding methods for boundary features to overcome these limitations. Furthermore, the model's adaptability to different scenes and ever-changing remote sensing images can be improved by incorporating incremental learning [65], [66] and domain adaptation techniques [67].

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